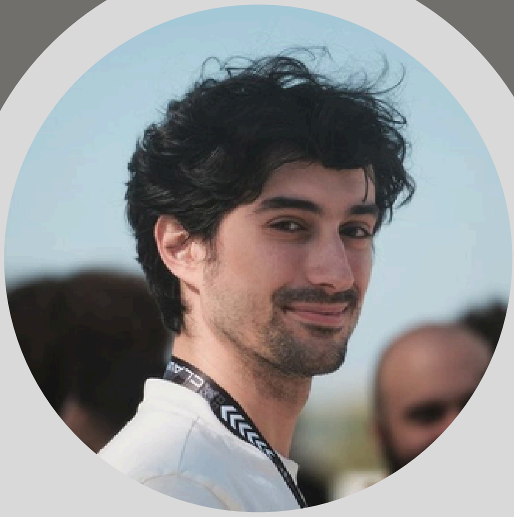
RICCARDO ZUCCHELLI

Network & Infrastructure Security



EDUCATION

2024-2026
MASTER IN SOFTWARE ENGINEERING - SAPIENZA

- Expected to graduate with full grades cum laude

2021 - 2024
BACHELOR IN ENGINEERING IN COMPUTER SCIENCE - SAPIENZA

- Graduated with full grades cum laude

SKILLS

- Proxmox virtualization & VM/ container mgmt
- Linux server administration (Debian, Ubuntu)
- Network security & firewalling
- P4 programmable switches
- Self-hosted services (reverse proxy, DNS)
- Monitoring, backups & recovery planning
- Troubleshooting & system maintenance
- Team collaboration in cross-functional projects

LANGUAGES

- English - C1
- Italian - Native

ACHIEVEMENTS

- PolyU International Research Summer School (Hong Kong, 2026)
- SPACEHACK Leonardo Hackathon (Rome, 2026)
- BSA Cryptocurrency Hackathon (Lausanne, 2025)

CONTACT

+39 3892433167
riccardo.zucchelli@gmail.com
Roma, 00193
github.com/rickythewoof

PROFILE INFO

Recent Computer Science graduate, specialising in security and networking, with practical experience in building and maintaining highly secure IT systems including personal infrastructure. Currently researching programmable P4 switching, combining routing and machine learning.

EXPERIENCE

JUNIOR CONSULTANT 2024–2026
SOLVIS S.R.L.

Developed embedded IoT firmware in Berry/C, implementing MQTT stack architecture and using it with Node-RED for orchestration, and device emulation

IT MANAGER
STUDIO MEDICO BASTIANELLI 2021–2024

Maintenance of critical IT/medical infrastructure, with ACLs to handle critical medical data

PROJECTS

SECURE SELF-HOSTED INFRASTRUCTURE

Proxmox VE virtualization, Portainer orchestration. Nginx as reverse proxy with TLS management. Applied network segmentation via Tailscale VPN for internal-only services and hardened public-facing endpoints. Grafana for monitoring, Crowdsec for blocking threats.
Some hosted services are Home automation (Home Assistant), photo management (Immich)

SDN LOAD BALANCING

Designed an SDN load-balancing controller in POX, combining connection-tracking-based session affinity with a capacity-weighted least-load algorithm to distribute new flows transparently across backend servers.

MACHINE-LEARNING-DRIVEN FORWARDING ON P4

Designed core mechanisms, extending the IPA paradigm to run on PISA architecture. Validated experimentally on BMv2 over the GÉANT topology